Axcel®

CEFUROXIME-500mg TABLETS

MAL21036034AZ

COMPOSITION:
Each tablet contains
Cefuroxime Axetil
equivalent to Cefuroxime 500mg

PRESENTATION:
White to off white, biconvex, capsule shape tablet with a scoreline on one side and B032 on other side.

INDICATIONS:
Cefuroxime axetil is an oral prodrug of the bactericidal cephalosporin antibiotic cefuroxime, which is resistant to most β-lactamases and is active against a wide range of Gram-positive and Gram-negative organisms. It is indicated for the treatment of infections caused by susceptible bacteria. Susceptibility to Cefuroxime axetil will vary with geography and time and local susceptibility data should be consulted where available (See Pharmacological properties, Pharmacodynamics). This indicated for the treatment of infections caused by sensitive bacteria which are including :

- Upper respiratory tract infections for example; ear, nose and throat infections, such as otitis media, sinusitis, tonsillitis and pharyngitis.
- Lower respiratory tract infections for example; pneumonia, acute bronchitis, and acute exacerbations of chronic bronchitis.
- Genito-urinary tract infections for example; pyelonephritis, cystitis and urethritis.
- Skin and soft tissue infections for example; furunculosis, pyoderma and impetigo.
- Gonorrhoea, acute uncomplicated gonococcal urethritis, and cervicitis.

Cefuroxime is also available as the sodium salt for parenteral administration. This permits the use of sequential therapy with the same antibiotic, when a change from parenteral to oral therapy is clinically indicated. Where appropriate, Axcel Cefuroxime-500mg Tablet is effective when used following initial parenteral Vaxcel Cefuroxime Injection (cefuroxime sodium) in the treatment of pneumonia and acute exacerbations of chronic bronchitis.

PHARMACOLOGY:
Mechanism of Action:
Cefuroxime axetil undergoes hydrolysis by esterase enzymes to the active antibiotic, cefuroxime. Cefuroxime inhibits bacterial cell wall synthesis following attachment to penicillin binding proteins (PBPs). This results in the interruption of cell wall (peptidoglycan) biosynthesis, which leads to bacterial cell lysis and death.

Mechanism of Resistance:
Bacterial resistance to cefuroxime may be due to one or more of the following mechanisms:

- Hydrolysis by beta-lactamases; including (but not limited to) by extended-spectrum beta-lactamases (ESBLs), and AmpC enzymes that may be induced or stably derepressed in certain aerobic Gram-negative bacteria species.
- Reduced affinity of penicillin-binding proteins for cefuroxime.
- Outer membrane impermeability, which restricts access of cefuroxime to penicillin binding proteins in Gram-negative bacteria.
- Bacterial efflux pumps.

Organisms that have acquired resistance to other injectable cephalosporins are expected to be resistant to cefuroxime. Depending on the mechanism of resistance, organisms with acquired resistance to penicillins may demonstrate reduced susceptibility or resistance to cefuroxime. The prevalence of acquired resistance is geographically and time dependent and for select species may be very high. Local information on resistance is desirable, particularly when treating severe infections. Cefuroxime is usually active against the following organisms *in vitro*. *In vitro* susceptibility of micro-organisms to Cefuroxime where clinical efficacy of cefuroxime axetil has been demonstrated in clinical trials this is indicated with an asterisk (*).

Commonly susceptible species
Gram-positive aerobes: *Staphylococcus aureus* (methicillin susceptible)*, Coagulase negative staphylococcus (methicillin susceptible), *Streptococcus pyogenes*, Beta-hemolytic streptococci. Gram-negative aerobes: *Haemophilus influenzae** including ampicillin resistant strains, *Haemophilus parainfluenzae**, *Moraxella catarrhalis**, *Neisseria gonorrhoea** including penicillinase and non-penicillinase producing strains . Gram-Positive Anaerobes: Peptostreptococcus spp., Propionibacterium spp. Spirochaetes: *Borrelia burgdorferi**.
Microorganisms for which acquired resistance may be a problem
Gram-positive aerobes: *Streptococcus pneumoniae**. Gram-negative aerobes: Citrobacter spp. not including *C. freundii*, Enterobacter spp. not including *E. aerogenes* and *E. cloacae*, *Escherichia coli**, Klebsiella spp. including *Klebsiella pneumoniae**, *Proteus mirabilis*, Proteus spp. not including *P. penneri* and *P. vulgaris*, Providencia spp. Gram-positive anaerobes: Clostridium spp. not including *C. difficile*. Gram-negative anaerobes: Bacteroides spp. not including *B. fragilis*, Fusobacterium spp.
Inherently resistant microorganisms
Gram-Positive Aerobes: Enterococcus spp. including *E. faecalis* and *E. faecium*, *Listeria monocytogenes*. Gram-negative aerobes: Acinetobacter spp., Burkholderia cepacia, Campylobacter spp., *Citrobacter freundii*, *Enterobacter aerogenes*, *Enterobacter cloacae*, *Morganella morganii*, *Proteus penneri*, *Proteus vulgaris*, Pseudomonas spp. including *Pseudomonas aeruginosa*, Serratia spp., *Stenotrophomonas maltophilia*. Gram-Positive Anaerobes: *Clostridium difficile*. Gram-negative anaerobes: *Bacteroides fragilis* Others: Chlamydia species, Mycoplasma species, Legionella species.

Pharmacokinetics:
Cefuroxime axetil is slowly absorbed from the gastrointestinal tract after oral administration and rapidly hydrolysed in the intestinal mucosa and blood to release cefuroxime into the circulation. Optimum absorption occurs when it is administered shortly after a meal. Peak serum level (2.1mg/l for a 125mg dose, 4.1mg/l for a 250mg dose, 7.0mg/l for a 500mg dose and 13.6mg/l for a 1g dose) occur approximately two to three hours after dosing when taken after food, unlike iv dosing which peaks immediately. The absorption of cefuroxime from the suspension is more prolonged compared with tablets, leading to later, lower peak serum levels and slightly reduced systemic bioavailability (4-17% less). Post peak levels, the serum half-life is between 1 and 1.5 hours. Protein binding has been variously stated as 33-50% depending on the methodology used. Cefuroxime is not metabolised and is excreted by glomerular filtration and tubular secretion. Concurrent administration of probenecid increases the area under the mean serum concentrations time curve by 50%. Serum levels of cefuroxime are reduced by dialysis.

Renal impairment:
Cefuroxime pharmacokinetics have been investigated in patients with various degrees of renal impairment. Cefuroxime elimination half-life increases with decrease in renal function which serves as the basis for dosage adjustment recommendations in this group of patients (See Dosage and Administration). In patients undergoing haemodialysis, at least 60% of the total amount of cefuroxime present in the body at the start of dialysis will be removed during a 4-hour dialysis period. Therefore, an additional single dose of cefuroxime should be administered following the completion of haemodialysis.

DOSAGE & ADMINISTRATION:
For oral administration only.

The usual course of therapy is seven days (range 5 to 10 days). Cefuroxime axetil should be taken after food for optimum absorption.

Dosage in adults:	
- Most infections	250mg twice daily
- Urinary tract infections	250mg twice daily
- Mild to moderate lower respiratory tract infections e.g. bronchitis	250mg twice daily
- More severe lower respiratory tract infections, or if pneumonia is suspected	500mg twice daily
- Pyelonephritis	250mg twice daily
- Uncomplicated gonorrhoea	Single dose of 1g

Sequential therapy:
Pneumonia:
1.5g cefuroxime injection (cefuroxime sodium) twice daily (given i.v. or i.m.) for 48 to 72 hours, followed by 500mg twice daily Axcel Cefuroxime Tablet (cefuroxime axetil) oral therapy for 7 to 10 days.

Acute exacerbations of chronic bronchitis:
750mg cefuroxime injection (cefuroxime sodium) twice daily (given i.v. or i.m.) for 48 to 72 hours, followed by 500mg twice daily Axcel Cefuroxime Tablet (cefuroxime axetil) oral therapy for 5 to 10 days. Duration of both parenteral and oral therapy is determined by the severity of the infection and the clinical status of the patient.

Dosage in children:
Most infections: 125mg (1 x 125mg tablet) twice daily.

Children with otitis media or, where appropriate, with more severe infections: 250mg (1 x 250mg tablet or 2 x 125mg tablets) twice daily.

There is no experience of using Axcel Cefuroxime Tablet in children under the age of 3 months.

Renal impairment:
Cefuroxime is primarily excreted by kidneys. In patients with markedly impaired renal function it is recommended that the dosage of cefuroxime be reduced to compensate for its slower excretion (see the table below).

Creatinine Clearance	$T_{1/2}$ (hours)	Recommended Dosage
≥ 30ml/min	1.4 - 2.4	No dose adjustment necessary standard dose of 125mg to 500mg given twice daily.
10-29ml/min	4.6	Standard individual dose given every 24 hours.
<10ml/min	16.8	Standard individual dose given every 48 hours.
During haemodialysis	2 – 4	A single additional standard individual dose should be given at the end of each dialysis.

CONTRAINDICATIONS:
a) Hypersensitivity to cefuroxime or to any of the excipients listed in the package insert.
b) Patients with known hypersensitivity to cephalosporin antibiotics.
c) History of severe hypersensitivity (e.g. anaphylactic reaction) to any other type of beta lactam antibacterial agent (penicillins, monobactams and carbapenems).

PRECAUTIONS:
Special care is indicated in patient who have experienced an allergic reaction to penicillins or other beta-lactams. Prolonged used of Axcel Cefuroxime-500mg Tablet may result in the overgrowth of other non-susceptible organism such as enterococci and *Clostridium difficile* which may require interruption of treatment. *Pseudomembranous colitis* has been reported with the use of broad-spectrum antibiotics, therefore

it's important to consider its diagnosis in patients who develop serious diarrhoea during or after antibiotic use. With a sequential therapy regime the timing of change to oral therapy is determined by severity of the infection, clinical status of the patient and susceptibility of the pathogens involved. If there is no clinical improvement within 72 hours, then the parenteral course of treatment must be continued. Please refer to the relevant prescribing information for cefuroxime sodium before initiating sequential therapy.

WARNINGS

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with Axcel Cefuroxime-500mg Tablet, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, Axcel Cefuroxime-500mg Tablet must be discontinued immediately and appropriate alternative therapy instituted.

Use in Pregnancy and Lactation:

There is no experimental evidence of embryopathic or teratogenic effects which refer to cefuroxime axetil, but it should be administered with caution during the early months of pregnancy. Cefuroxime is excreted in human milk, and therefore caution should be exercised when cefuroxime axetil is administered to a nursing mother.

SIDE EFFECTS:

Adverse reaction to Cefuroxime Axetil is generally mild and transient in nature. The incidence of adverse reaction associated with Cefuroxime Axetil may vary according to the indication. The following side effects have been observed:

Infections and infestations: overgrowth of candida.

Blood and lymphatic system disorders: Eosinophilia, Positive Coombs test, thrombocytopenia, leukopenia (sometimes profound) and Haemolytic anaemia. Cephalosporin tend be absorbed onto the surface of red cells membranes and react with antibodies directed against the drug to produce a positive Coombs test (interfere with cross-matching of blood) and very rarely haemolytic anaemia.

Immune system disorders: Hypersensitivity reactions including skin rashes, urticaria, pruritus, drug fever, serum sickness, anaphylaxis.

Nervous system disorders: Headache, dizziness.

Gastrointestinal disorders: Gastrointestinal disturbances including diarrhoea, nausea, abdominal pain, vomiting, and *Pseudomembranous colitis*.

Hepatobiliary disorders: Transient increases of hepatic enzyme levels, [ALT(SGPT), AST (SGOT), LDH], jaundice (predominantly cholestatic), and hepatitis.

Skin and subcutaneous tissue disorders: Erythermamultiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (exanthematic necrolysis), see on immune system disorders.

Axcel Cefuroxime-500mg Tablet may cause severe allergy and serious skin reactions. Stop using Axcel Cefuroxime-500mg Tablet and seek medical assistance immediately if you experience any of the following symptoms:

•skin reddening, blisters, rash, fever, sore throat or eye irritation.

Effects on ability to drive and use machines:

This medicine may cause dizziness, patient should be more careful while driving or operating machinery.

DRUG INTERACTIONS:

Drug which reduce gastric acidity may result in a lower bioavailability of Cefuroxime Axetil compared with that of the fasting state and tend to cancel the effect of enhanced post-prandial absorption. Similar with other antibiotics, Axcel Cefuroxime-500mg Tablet may affect the gut flora, leading to lower oestrogen reabsorption and reduced efficacy of combined oral contraceptives. As a false negative result may occur in the ferricyanide test, it is recommended that either the glucose oxidase or hexokinase methods are used to determine blood / plasma glucose levels in patients receiving cefuroxime axetil. This antibiotic does not interfere in the alkaline picrate assay for creatinine.

OVERDOSAGE AND TREATMENT:

Overdosage of cephalosporins can cause cerebral irritation leading to convulsions. Serum levels of cefuroxime can be reduced by haemodialysis and peritoneal dialysis.

STORAGE:

Store below 30^{O}C. Protect from light.

KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK

PACK QUANTITIES

Available in blister pack of 1 x 10's and 10 x 10's.
Not all pack sizes may be marketed.

Further information can be obtained from pharmacist, physician or the manufacturer.

Product Registration Holder & Manufactured By :
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